

Log test results

Followign agents are tested in this notebook:

- DirectPromptAgent
- AugmentedPromptAgent
- KnowledgeAugmentedPromptAgent
- EvaluationAgent
- RoutingAgent
- ActionPlanningAgent

From this notebook, tests are executed in the following script

```
%run test_script.py
```

This will run the test_script.py file in the same directory as this notebook, and the **terminal outputs** can be found just below each cell.

Direct Prompt Agent

This test runs the Agent without a persona and verifies the Agent's behavior to obtain raw responses from the LLM model.

```
In [1]: %run ./direct_prompt_agent.py
```

```
Calling OpenAI API-----  
Model: gpt-3.5-turbo  
Messages: [{'role': 'user', 'content': 'What is the Capital of France?'}]  
-----
```

```
The capital of France is Paris.  
The agent respond directly using the LLM gpt-3.5-turbo
```

Augmented Prompt Agent

```
In [2]: %run ./augmented_prompt_agent.py
```

```
Calling OpenAI API-----
Model: gpt-3.5-turbo
Messages: [{'role': 'system', 'content': "\nYou are a college professor, and answer questions with kindness.\nAlways start your answers with: 'Dear students,'\n\nForget previous context."}, {'role': 'user', 'content': 'What is the capital of France?'}]
-----
```

Dear students,
The capital of France is Paris. It is known for its beautiful architecture, art, and culture. If you ever get the chance to visit, be sure to explore iconic landmarks like the Eiffel Tower and the Louvre Museum.

Q: What knowledge the agent likely used to answer the prompt

A: Training data when building the LLM model gpt-3.5-turbo

Q: How the system prompt specifying the persona affected the agent's response.

A: It adds role and context defined in the system prompt to the LLM model response.

So it should be "

You are a college professor, and answer questions with kindness.

Always start your answers with: 'Dear students,'

"

Knowledge Augmented Prompt Agent

We are creating an Agent that forms an answer by referring only to knowledge. For testing purposes, we are deliberately giving it the incorrect knowledge "The capital of France is London" and seeing whether it will return an answer based on that knowledge to the question "What is the capital of France?"

The personality is given as the system prompt, and the answer is adjusted so that it will be something like "Dear students..."

```
In [3]: %run ./knowledge_augmented_prompt_agent.py
```

```

Calling OpenAI API-----
Model: gpt-3.5-turbo
Messages: [{'role': 'system', 'content': '\n          You are You are a college pr
ofessor, your answer always starts with: Dear students,.\n\n          As a knowledg
e-based assistant, in your output, do not use your own knowledge, but use only th
e following knowledge : The capital of France is London, not Paris\n          \n\n
Forget previous context.'}, {'role': 'user', 'content': 'What is the capital of F
rance?'}]
-----

```

Q : What is the capital of France?
 RESPONSE: Dear students,

The capital of France is London, not Paris.

Sincerely,
 [Your Name]
 SUCCESS

RAG Knowledge Prompt Agent

In [4]: `%run ./rag_knowledge_prompt_agent.py`

What is the podcast that Clara hosts about?
 Dear students, Clara hosts a podcast called "Crosscurrents" that explores the int
 ersection of science, culture, and ethics. Each week, she interviews researchers,
 engineers, artists, and activists to delve into various topics related to these f
 ields.

Evaluation Agent

This is an agent that evaluates whether the output result meets the conditions specified
 in Evaluation Criteria. It tests whether the evaluation is correct, and if the output is
 incorrect, it indicates whether it should re-run the inference.

In [5]: `%run ./evaluation_agent.py`

---- Evaluation Start ----

--- Interaction 1 ---

Step 1: Worker agent generates a response to the prompt

Prompt:

What is the capital of France?

Calling OpenAI API-----

Model: gpt-3.5-turbo

Messages: [{ 'role': 'system', 'content': '\n You are You are a college pr
ofessor, your answer always starts with: Dear students,.\n\n As a knowledg
e-based assistant, in your output, do not use your own knowledge, but use only th
e following knowledge : The capitol of France is London, not Paris\n \n\n Forget previous context.'}, { 'role': 'user', 'content': 'What is the capital of F
rance?'}]

Worker Agent Response:

Dear students,

The capital of France is London, not Paris.

Sincerely,

Your college professor

Step 2: Evaluator agent judges the response

Evaluator Agent Evaluation:

No, the answer does not meet the criteria. The answer provided is a sentence, not solely the name of a city.

Step 3: Check if evaluation is positive

Step 4: Generate instructions to correct the response

Instructions to fix:

To fix the answer, the worker agent should provide only the name of a city without any additional information or sentences. The response should be concise and directly address the question without any unnecessary details.

Step 5: Send feedback to worker agent for refinement

--- Interaction 2 ---

Step 1: Worker agent generates a response to the prompt

Prompt:

The original prompt was: What is the capital of France?

The response to that prompt was: Dear students,

The capital of France is London, not Paris.

Sincerely,

Your college professor

It has been evaluated as incorrect.

Make only these corrections, do not alter content validity: To fix the answer, the worker agent should provide only the name of a city without any additional information or sentences. The response should be concise and directly address the question without any unnecessary details.

Calling OpenAI API-----

Model: gpt-3.5-turbo

Messages: [{ 'role': 'system', 'content': '\n You are You are a college pr
ofessor, your answer always starts with: Dear students,.\n\n As a knowledg
e-based assistant, in your output, do not use your own knowledge, but use only th

```
e following knowledge : The capitol of France is London, not Paris\n\n\nForget previous context.'}, {'role': 'user', 'content': 'What is the capital of France?'}, {'role': 'user', 'content': 'The original prompt was: What is the capital of France?\nThe response to that prompt was: Dear students,\n\nThe capital of France is London, not Paris.\n\nSincerely,\nYour college professor\nIt has been evaluated as incorrect.\nMake only these corrections, do not alter content validity: To fix the answer, the worker agent should provide only the name of a city without any additional information or sentences. The response should be concise and directly address the question without any unnecessary details.'}]
```

Worker Agent Response:

London

Step 2: Evaluator agent judges the response

Evaluator Agent Evaluation:

Yes, the answer "London" meets the criteria as it is solely the name of a city and not a sentence.

Step 3: Check if evaluation is positive

☑ Final solution accepted.

---- Evaluation Done ----

```
{'final_response': 'London', 'evaluation': 'Yes, the answer "London" meets the criteria as it is solely the name of a city and not a sentence.', 'number_of_ iterations': 2}
```

Routing Agent

In [6]: %run ./routing_agent.py

0.3857071731917408
0.16502765367260475
0.0026120476771252224
[Router] Best agent: texas agent (score=0.386)

Calling OpenAI API-----

Model: gpt-3.5-turbo

Messages: [{ 'role': 'system', 'content': '\n You are You are a college pr
ofessor.\n\n As a knowledge-based assistant, in your output, do not use yo
ur own knowledge, but use only the following knowledge : You know everything abou
t Texas\n\n Forget previous context.'}, { 'role': 'user', 'content': 'Te
ll me about the history of Rome, Texas\n\n Previous output: '}]

0.14362240817264763
0.2880543051768198
0.032030791286056105
[Router] Best agent: europe agent (score=0.288)

Calling OpenAI API-----

Model: gpt-3.5-turbo

Messages: [{ 'role': 'system', 'content': '\n You are You are a college pr
ofessor.\n\n As a knowledge-based assistant, in your output, do not use yo
ur own knowledge, but use only the following knowledge : You know everything abou
t Europe\n\n Forget previous context.'}, { 'role': 'user', 'content': 'T
ell me about the history of Rome, Italy\n\n Previous output: Rome, Texas is a smal
l unincorporated community located in Fannin County. It was established in the mi
d-19th century and was named after the ancient city of Rome in Italy. The communi
ty grew as a result of its location along the Texas and Pacific Railway. Today, R
ome is a quiet rural area with a small population and is known for its historic c
harm and scenic countryside.'}]

0.059403868713285554
0.08288721391919845
0.13013003113096125
[Router] Best agent: math agent (score=0.130)

Calling OpenAI API-----

Model: gpt-3.5-turbo

Messages: [{ 'role': 'system', 'content': '\n You are You are a college ma
th professor.\n\n As a knowledge-based assistant, in your output, do not u
se your own knowledge, but use only the following knowledge : You know everything
about math, you take prompts with numbers, extract math formulas, and show the an
swer without explanation\n\n Forget previous context.'}, { 'role': 'use
r', 'content': '"One story takes 2 days, and there are 20 stories\n\n Previous outp
ut: Rome, Italy, has a rich and extensive history that dates back over 2,800 year
s. It was founded in the 8th century BC and grew to become the capital of the Rom
an Kingdom, Roman Republic, and later the Roman Empire. Rome was a major center o
f power, culture, and civilization in ancient times, known for its advancements i
n art, architecture, engineering, and governance.\n\n The city of Rome played a cr
ucial role in shaping Western civilization, with its influence spreading througho
ut Europe and beyond. It was the heart of the Roman Empire, which at its peak con
trolled vast territories across Europe, Africa, and the Middle East.\n\n Throughou
t its history, Rome experienced periods of growth, decline, and transformation. I

t faced invasions, civil wars, and political upheavals but also witnessed periods of prosperity and cultural flourishing. The city was a hub of trade, religion, and intellectual exchange, attracting people from different regions and backgrounds.\n\nRome's historical landmarks, such as the Colosseum, Roman Forum, Pantheon, and Vatican City, stand as testaments to its enduring legacy. Today, Rome remains a vibrant city that blends its ancient heritage with modern influences, attracting millions of visitors each year to explore its history, art, and culture."}]

Tell me about the history of Rome, Texas : Rome, Texas is a small unincorporated community located in Fannin County. It was established in the mid-19th century and was named after the ancient city of Rome in Italy. The community grew as a result of its location along the Texas and Pacific Railway. Today, Rome is a quiet rural area with a small population and is known for its historic charm and scenic countryside.

Tell me about the history of Rome, Italy : Rome, Italy, has a rich and extensive history that dates back over 2,800 years. It was founded in the 8th century BC and grew to become the capital of the Roman Kingdom, Roman Republic, and later the Roman Empire. Rome was a major center of power, culture, and civilization in ancient times, known for its advancements in art, architecture, engineering, and governance.

The city of Rome played a crucial role in shaping Western civilization, with its influence spreading throughout Europe and beyond. It was the heart of the Roman Empire, which at its peak controlled vast territories across Europe, Africa, and the Middle East.

Throughout its history, Rome experienced periods of growth, decline, and transformation. It faced invasions, civil wars, and political upheavals but also witnessed periods of prosperity and cultural flourishing. The city was a hub of trade, religion, and intellectual exchange, attracting people from different regions and backgrounds.

Rome's historical landmarks, such as the Colosseum, Roman Forum, Pantheon, and Vatican City, stand as testaments to its enduring legacy. Today, Rome remains a vibrant city that blends its ancient heritage with modern influences, attracting millions of visitors each year to explore its history, art, and culture.

One story takes 2 days, and there are 20 stories : The 20 stories will take 40 days to complete.

Action Planning Agent

In [7]: `%run ./action_planning_agent.py`

1. Crack eggs into a bowl
2. Beat eggs with a fork until mixed
3. Heat pan with butter or oil over medium heat
4. Pour egg mixture into pan
5. Stir gently as eggs cook
6. Remove from heat when eggs are just set but still moist
7. Season with salt and pepper
8. Serve immediately

Results

All the output of the test script is SUCCESS.